The Ramification Index:

RAM Prices, Oligopoly Cycles, and the Downstream Consequences of Semiconductor Pricing as an Economic Signal

Version 4: The Contested Oligopoly — CXMT, the Fourth Firm, and the Geopolitical Ramification of Memory

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August 2026

Abstract

[Append the following to the end of the v3 abstract.]

This fourth version extends the third in four ways, all organised around a single event: the July 2026 public listing of ChangXin Memory Technologies (CXMT), a state-backed Chinese producer and the first credible fourth entrant into a DRAM market the prior versions treated as a stable three-firm oligopoly. (1) We document the entry of a fourth manufacturer holding roughly 7.7–9% of global DRAM and its record STAR Market debut, and show why a state-backed firm optimising for national self-sufficiency rather than oligopoly pricing discipline breaks the coordination assumption underpinning the index's supply-side interpretation. (2) We introduce a *second bifurcation*: having split the index by product (commodity versus AI/HBM), we now split it by geography, because U.S. export controls firewall a China-domestic memory market from the rest of the world, causing the single price series to branch into two. (3) We add a fifth interpretive regime—*subsidised-supply divergence*—in which the commodity sub-index turns negative not from demand collapse but from subsidised capacity expansion, and we warn that this regime is observationally the mirror image of a recession signal, creating a false-positive risk the index must actively guard against. (4) We specify how to admit CXMT to the index without importing the policy premium embedded in its equity, whose 466% debut priced a geopolitical option rather than near-term fundamentals. The fourth firm does not dissolve the oligopoly; it makes the oligopoly *contested*, and a contested oligopoly ramifies differently.

[Intro roadmap — replace the "third version" sentence:]
This fourth version extends the framework to a structural break the earlier versions could not have anticipated: the arrival of a fourth, state-backed, publicly traded manufacturer. Section 13 documents the CXMT listing, formalises the "contested oligopoly," introduces

the geographic bifurcation, adds a fifth interpretive regime, and specifies the de-politicisation adjustment required to admit CXMT to the index.

13 The Fourth Firm: CXMT and the Contested Oligopoly

The prior versions of this paper rest on a load-bearing assumption stated plainly in the abstract: three firms—Samsung, SK hynix, and Micron—control roughly 95% of the world's dynamic random-access memory, and their coordinated response to the AI upcycle is what makes DRAM pricing a clean supply-side signal. On 27 July 2026 that assumption acquired an asterisk.

13.1 The event

ChangXin Memory Technologies (CXMT), founded with state backing in Hefei in 2016, listed on the Shanghai Stock Exchange's STAR Market in the largest semiconductor IPO in that board's history. It priced 6.688 billion shares at ¥8.66, raising ¥57.92 billion (approximately US$8.6 billion), with proceeds reaching ¥66.61 billion if the over-allotment is exercised [1, 7]. On debut the stock rose about 466%, closing at ¥49 after an intraday high of ¥55.03, lifting CXMT's market capitalisation to roughly ¥3.3 trillion (about US$488 billion), surpassing Industrial and Commercial Bank of China to make it, briefly, the most valuable company listed on any mainland Chinese exchange [1, 9]. Its prospectus reported a 7.67% share of the 2025 global DRAM market [2]; independent trackers place it fourth worldwide at roughly 8% of 2025 shipments, rising to about 9% in the first quarter of 2026, behind Samsung (36%), SK hynix (29%), and Micron (24%) [3]. The oligopoly is intact (Figure 1a); it is no longer uncontested.

13.2 A different objective function

The analytically important fact is not CXMT's size but its *incentive structure*. The three incumbents are profit-maximising, dividend-paying, and capacity-disciplined; their reallocation of wafer starts toward high-margin HBM is precisely the behaviour that produced the supply-side divergence of 2025–2026 (Section §13 presupposes the v2 argument). CXMT operates under a different mandate. It is an instrument of a national self-sufficiency programme, capitalised in part by strategic investors including Alibaba, Xiaomi, and Meituan [8], and its objective function weights domestic supply security alongside—at times above—margin. An oligopolist that does not need to maximise profit is not bound by the oligopoly's pricing discipline. This is the mechanism by which a fourth firm alters the signal: the index reads DRAM prices as revealed oligopoly behaviour, and a participant playing a different game contaminates that read.

For now the contamination is latent. The memory market remains tight, and CXMT has exercised, not surrendered, pricing power—reportedly refusing discount requests from large domestic buyers [6]. TrendForce expects price increases to continue through the end of 2027, and notes that customers seeking to diversify their supplier base structurally benefit

CXMT [6]. The regime shift this section formalises is therefore prospective, not yet realised—which is exactly why the index needs to be able to recognise it *before* it arrives.

13.3 The asymmetry: market capitalisation is not firepower

A naive reading of the debut—China's most valuable listed company, a half-trillion-dollar memory champion—overstates the threat. The relevant metric for a capacity war is investment, and here the fourth firm remains a minnow (Figure 1b). Samsung's FY2026 planned investment exceeds US$73 billion; Micron's FY2026 capital expenditure exceeds US$25 billion; SK hynix raised US$26.5 billion in a Nasdaq listing days before CXMT's subscription window opened [4]. The combined incumbent commitment is on the order of fourteen times CXMT's IPO haul. The equity market is not pricing near-term fundamentals: at the offer price CXMT's trailing P/E was roughly 309× against a forward P/E near 5.8× [8]. What the debut priced was a policy option—the strategic value of a domestic champion—rather than a cash-flow forecast. *This distinction is the crux of the incorporation problem in §13.6: CXMT's equity is a geopolitical instrument first and a memory stock second.*

13.4 The second bifurcation: geography

Version 2 bifurcated the index by product, separating a commodity sub-index R_t^C from an AI/HBM sub-index R_t^{AI} to capture the structural decoupling of HBM pricing. The CXMT event forces a second, orthogonal bifurcation—by geography. CXMT sits on the U.S. Entity/defence blacklists and operates under export controls that restrict its access to leading-edge fabrication tools [3, 10]; conversely, Apple has reportedly tested CXMT DRAM for devices sold in China and sought U.S. government permission to broaden that use [2, 10]. The result is a partial firewall: a China-domestic memory market increasingly supplied by CXMT, and a rest-of-world market still governed by the three incumbents. We therefore split the price signal:

$$R_t \;\longrightarrow\; \big(R_t^{\mathrm{dom}},\, R_t^{\mathrm{RoW}}\big), \tag{1}$$

where R_t^{dom} tracks pricing inside the firewalled Chinese market and R_t^{RoW} tracks the rest. The two need not co-move, and their divergence is itself informative: a widening gap measures the degree of market fragmentation, and hence the erosion of the single global oligopoly price that made the original index legible. Here the paper's titular conceit becomes literal. In algebraic number theory the ramification index counts how a prime splits as it passes into a field extension; the memory market has just acquired a new prime, and the single price series is beginning to *ramify* geographically.

13.5 A fifth interpretive regime

Version 1 gave three interpretive cases and Version 2 added a fourth—the stagflationary "supply-side divergence" in which $R_t^C \gg 0$ coexists with weakening macro indicators. The contested oligopoly requires a fifth. Suppose a subsidised entrant adds capacity fast enough to soften commodity DRAM prices *without* any contraction in end demand. Then R_t^C turns negative for a reason that has nothing to do with the business cycle:

#	Condition	Economic meaning	Ver.
1	$R_t \ll 0$	Demand-side contraction (recession signal)	v1
2	$R_t \approx 0$	Equilibrium	v1
3	$R_t \gg 0$	Healthy demand growth (expansion signal)	v1
4	$R_t^C \gg 0$ **and** macro weakening	Supply-side divergence (stagflationary; oligopoly reallocation to AI)	v2
5	$R_t^C < 0$ **and** volumes rising **and** no demand weakness	Subsidised-supply divergence (contested oligopoly; incumbent margin compression — *not* a recession)	v4

Table 1: The interpretive framework, extended. Cases 1 and 5 share a sign but carry opposite meanings; distinguishing them is the central measurement problem introduced by the fourth firm.

$R_t^C < 0$ **and** shipment volumes rising **and** no demand weakness $\Rightarrow$ **subsidised-supply divergence.**

The danger is observational. A negative commodity sub-index is, in Version 1, the paper's cleanest recession signal (case 1). In the contested-oligopoly regime an identical reading means the opposite: not demand destruction but supply abundance manufactured by policy, whose downstream effect is margin compression for incumbents, not economic contraction. An index that cannot distinguish case 1 from case 5 will emit a false recession signal precisely when a state-backed competitor succeeds. Disambiguation requires auxiliary series the price signal alone does not carry—shipment volumes, CXMT capacity additions, and the $R_t^{\text{dom}}/R_t^{\text{RoW}}$ spread. Table 1 consolidates all five regimes.

13.6 Incorporating CXMT into the index

Two rules follow. First, *weight by shipment share, not market capitalisation.* CXMT's $\sim 8\%$ shipment share, not its half-trillion-dollar valuation, is its weight in any supply-side aggregate; folding a policy-inflated equity into the index would import exactly the noise the index exists to filter. Second, *treat CXMT's equity as a separate indicator, not a component.* Its stock is best read as a "policy put"—a barometer of Beijing's commitment to memory self-sufficiency and of the market's expectations for export-control escalation—and tracked alongside the index rather than inside it. The bipartisan U.S. congressional inquiry opened the day after the debut [4, 5] is a movement in that policy variable, not in the DRAM supply curve, and the index should not confuse the two.

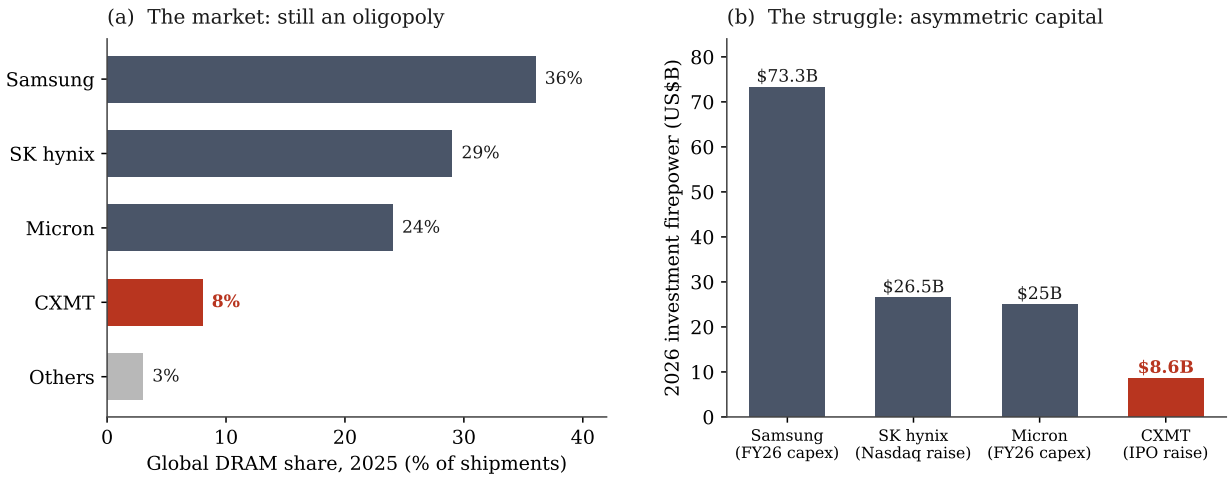


Figure 1: The contested oligopoly. **(a)** Global DRAM share, 2025, by shipments: the three incumbents still hold roughly 89%, with CXMT a distant but established fourth (Counterpoint estimates; the CXMT prospectus reports 7.67% on a Q4-2025 revenue basis). **(b)** 2026 investment firepower: CXMT's record IPO haul is dwarfed by incumbent capital commitments. Note that the panel mixes annual capital-expenditure plans (Samsung, Micron) with one-time equity raises (SK hynix Nasdaq listing, CXMT IPO); it is intended to convey order-of-magnitude asymmetry, not a like-for-like comparison. Sources: [1, 3, 4].

13.7 Summary

The fourth firm does not falsify the Ramification Index; it dates it. The three-firm oligopoly was a feature of a particular decade, and the index was built to read that structure. A state-backed, publicly traded, export-controlled fourth entrant introduces a second bifurcation (geographic), a fifth interpretive regime (subsidised-supply divergence), and a live false-signal risk (case 5 masquerading as case 1). The index remains a valid supply-side instrument; it now requires the auxiliary series to stay valid as the market ramifies.

References

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